

Session 8 :Functional Programming using reduce()

1.Maximum Subarray:

```
class Solution {  
    public int maxSubArray(int[] nums) {  
        int currentSum = nums[0];  
        int maxSum = nums[0];  
        for (int i = 1; i < nums.length; i++) {  
            currentSum = Math.max(nums[i], currentSum + nums[i]);  
            maxSum = Math.max(maxSum, currentSum);  
        }  
        return maxSum;  
    }  
}
```



2. Find the Highest Altitude:

```
class Solution {  
    public int largestAltitude(int[] gain) {  
        int altitude = 0;  
        int highest = 0;  
        for (int i = 0; i < gain.length; i++) {  
            altitude = altitude + gain[i];  
            if (altitude > highest) {  
                highest = altitude;  
            }  
        }  
        return highest;  
    }  
}
```

1732. Find the Highest Altitude

There is a biker going on a road trip. The road consists of $n + 1$ points of different altitudes. The biker starts at point 0 with an altitude of 0. You are given an array of n points of the road, where $points[i]$ represents the change in altitude between point i and point $i + 1$. Your task is to find the highest altitude of a point in the road.

Example 1:
Input: $gain = [-5, 1, 2, -4, -3]$
Output: 0
Explanation: The altitudes are $[0, -5, -4, -2, -5, -8]$. The highest is 0 at point 0.

Example 2:
Input: $gain = [-4, -2, -1, 3, -1, 4, -1, 4]$
Output: 6
Explanation: The altitudes are $[0, -4, -6, -7, -4, -1, 3, 2, 6, 3]$. The highest is 6 at point 9.

Constraints:

- $n == points.length$
- $1 \leq n \leq 1000$
- $-10^5 \leq points[i] \leq 10^5$

```

class Solution {
public:
    int largestAltitude(vector<int> &points) {
        int maxAltitude = 0;
        int currentAltitude = 0;
        for (int i = 0; i < points.size(); i++) {
            currentAltitude += points[i];
            maxAltitude = max(maxAltitude, currentAltitude);
        }
        return maxAltitude;
    }
};

```